```
> restart; # The propagation equation
> #  (df(r,t))/dt=(∂[u(t)g(r)])/∂t+(∂[u(t)g(r)])/∂r·∂r/∂t=c
> eq1 := g(r)·diff(u(t), t) + u(t)·diff(g(r), r)·c − c = 0;
```

$$eq1 := g(r)\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right) + u(t)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)\right)c - c = 0 \tag{1}$$

```
> iv1 := D[1](g)(0) = R
```

$$iv1 := \mathrm{D}(g)(0) = R \tag{2}$$

```
> dsolve({eq1, iv1}, g(r));#dsolve(eq1, u(t));
```

$$g(r) = \frac{c}{\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)} - \frac{\mathrm{e}^{-\frac{\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)r}{u(t)\,c}}\,R\,u(t)\,c}{\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)} \tag{3}$$

```
> #df(r,z,t)/dt=∂[u(t) H3 (r) H4 (z)]/∂t+∂[u(t) H3 (r) H4 (z)]/∂r·∂r/∂t+∂
    [u(t) H3 (r) H4 (z)]/∂z·∂z/∂t=c where  ∂z/∂r=∂z/∂t=0 and ∂r/∂t=c
> eq2 := H4(z)·diff(u(t)·H3(r), t) + u(t)·diff(H3(r)·H4(z), r)·c + diff(u(t)·H3(r)
    ·H4(z), z)·diff(z, t) − c = 0;
```

$$eq2 := H4(z)\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)H3(r) + u(t)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right)H4(z)\,c - c = 0 \tag{4}$$

```
> iv2 := D[1](H3)(0) = R;
```

$$iv2 := \mathrm{D}(H3)(0) = R \tag{5}$$

```
> dsolve(eq2, u(t)); dsolve({eq2, iv2}, H3(r)); solve({eq2, iv2}, H4(z));
```

$$u(t) = \frac{1}{H4(z)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right)} + \mathrm{e}^{-\frac{\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right)ct}{H3(r)}}_C1$$

$$H3(r) = \frac{c}{H4(z)\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)} - \frac{\mathrm{e}^{-\frac{\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)r}{u(t)\,c}}\,R\,u(t)\,c}{\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)} \tag{6}$$

```
> #H3 (r)  (∂u(t))/∂t+u(t)  (∂H3 (r))/∂r=c/(H4 (z) );
> eq3 := (d/dt u(t)) H3(r)/c + u(t) (d/dr H3(r)) = 1/H4(z);
```

$$eq3 := \frac{\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)H3(r)}{c} + u(t)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right) = \frac{1}{H4(z)} \tag{7}$$

```
> iv3 := D[1](g)(1/R) = R
```

$$iv3 := \mathrm{D}(g)\left(\frac{1}{R}\right) = R \tag{8}$$

```
>  dsolve(eq3, u(t)); dsolve(eq3, H3(r)); solve(eq3, H4(z));
```

$$u(t) = \frac{1}{H4(z)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right)} + \mathrm{e}^{-\frac{\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right)ct}{H3(r)}}_C1$$

$$H3(r) = \frac{c}{H4(z)\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)} + \mathrm{e}^{-\frac{\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right) r}{u(t)\,c}}\,_C1$$

$$\frac{c}{u(t)\left(\frac{\mathrm{d}}{\mathrm{d}r}\,H3(r)\right) c + \left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right) H3(r)} \tag{9}$$

>